Supplemental Material: Group-flip consensus on hypergraphs: exact stationary band at high coverage, exact one-dimensional containment, and an exploratory controlled test of the one-step containment hypothesis

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S0. Notation, labeling rules, and crosswalk to main-text pointers

Objects.. $H = (V, E)$, $|V| = n$; hypergraph distance $d(\cdot, \cdot)$ = minimum number of hyperedge steps (a step from i reaches any node sharing a hyperedge with i); ball $B_i = B_i(r) = \{j : d(i, j) \leq r\}$, $\beta_i = |B_i|$, β when uniform; coverage $q = \beta/n$; residual $R_i = V \setminus B_i$. Conditions: **R1** $\beta_i > n/2$ $\forall i$; **R2** $\mathrm{diam}_H(R_i) \leq r$ $\forall i$. Dynamics: each node fires at rate 1; a fire at i moves *all* nodes of color σ_i inside B_i to a target color drawn uniformly among the $Q - 1$ others. Observables: stationary majority fraction M; epoch counts K (fires between consecutive rotations) and T (physical cycle time). Containment: $c_G, \tilde{c}$ as in main-text Eq. (2), with $\tilde{c} = 1$ if all residuals empty.

Labeling rules.. *Theorem/Proposition/Lemma*: complete proof given here, except where the statement's own text explicitly says "proof sketch" or "status: validated numerically"; in those cases the label records the intended status and the proof gap is stated in the environment body, not hidden. *Conjecture*: numerically tested statement, no proof claimed. *Heuristic*: structural or scaling claim without a proof attempt. *Protocol*: measurement/statistical design, not a mathematical theorem. *Remark*: scope clarification; each Remark below is normative (it delimits exactly what is and is not claimed). Two environments are explicitly downgraded in this document: Proposition B.4 (wrapped regime, numerical only) and Proposition B.5 (sparse configuration model, conditional theorem with sketch); their labels are retained for cross-reference but their proof status is stated in the body.

Crosswalk.. Main-text pointers map as: "SM §A"=S1 (well-mixed), "SM §B"=S2 (geometry), "SM §C"=S3 (band), "SM §D"=S4 (containment), "SM §E"=S5 (1D), "SM §F"=S6 (protocols/validation), "SM §S6.5"=S6.5 (cluster dependence), "SM §G"=S7 (open), "SM §H"=S8 (numerical validation summary), numerical tables=S9. The well-mixed limit is S1/A; limitations are in S7.1.

Normalization rule.. All exact formulas use $q = \beta/n$. Any pipeline reporting an alternative coverage normalization must convert before comparison; no result in this SM mixes normalizations.

S1 (A). Well-mixed limit: count chain, pure death, consensus, dark horse

Lemma A.1 (Count-chain reduction)**.** *If $B_i = V$ for all i, the color-count vector $x = (x_1, \ldots, x_Q)$, $\sum_k x_k = n$, is Markov on the simplex with transitions $x \to x + x_k(e_l - e_k)$ at rate $x_k/(Q-1)$ for each ordered pair (k, l), $k \neq l$.*

Proof. A fire at a node of color k (total rate x_k, nodes fire at rate 1) flips the entire class k (since $B_i = V$) to the target l, chosen with probability $1/(Q-1)$. The kernel depends only on x, hence lumping configurations by counts is exact. □

Lemma A.2 (Pure death of live colors)**.** *Let L_t = number of colors with ≥ 1 node. L_t is pure-death on $\{1, \ldots, Q\}$ with $L \to L-1$ at rate $n(L-1)/(Q-1)$; $L = 1$ absorbing.*

Proof. Since every node carries a live color, $\sum_{k \text{ live}} x_k = n$ at all times. A fire from live class k decreases L iff the target is another live class, probability $(L-1)/(Q-1)$; summing over live k gives rate $n(L-1)/(Q-1)$. Targets that are absent colors merely relabel the firing class, leaving L unchanged; no event increases L. At $L = 1$ the rate is 0. □

Theorem A.3 (First-consensus time)**.** *Assume all Q colors are initially live; if only $L_0 \leq Q$ colors are live, the sum runs from $\ell = 2$ to L_0. In time units with node rate 1,*

$$E[T_{\text{cons}}] = \frac{Q-1}{n} \sum_{\ell=2}^{Q} \frac{1}{\ell - 1} = \frac{Q-1}{n} H_{Q-1} \quad \textit{(when } L_0 = Q\textit{)}.$$

For $Q = 2$: $E[T] = 1/n$.

Proof. Holding time at level ℓ is Exp with rate $n(\ell-1)/(Q-1)$ (Lemma A.2); sum the means over $\ell = Q, \ldots, 2$. For $Q = 2$ the first fire always merges the two classes. □

Remark A.3.1 (A.3.1). The harmonic form (not a telescoping $(Q-1)/(\ell(\ell-1))$ form) is the correct one for whole-class flips; any quoted telescoping prefactor corresponds to a different (pair-sampling) update and is not used here.

Theorem A.4 (Minority advantage, $Q = 2$)**.** *From $(a, n-a)$, the class of mass a wins the first consensus with probability $1 - a/n$.*

Proof. With $Q = 2$ the first fire decides: firing node in class a (prob. a/n) flips class a into the other color (class a loses); otherwise class a wins. □

Theorem A.5 (Dark horse, $Q = 3$, closed form)**.** *Assume $a, b > 0$ with $a + b = n$, colors $1, 2$ live with masses a, b, and color 3 absent. Let $D = a^2 + 3ab + b^2$. Then the probability P_k that color k wins the first consensus is $P_1 = b(a+b)/D$, $P_2 = a(a+b)/D$, $P_3 = ab/D$, with $P_1 + P_2 + P_3 = 1$. Note P_3 is the probability that the currently-absent color is regenerated and wins; P_1 and P_2 are win probabilities of the currently-live colors. $P_1 = \frac{b(a+b)}{D}$, $P_2 = \frac{a(a+b)}{D}$, $P_3 = \frac{ab}{D}$, and $P_1 + P_2 + P_3 = 1$.*

Proof. First-step analysis with $F = P_1(a, b \text{ live})$, $G = P_1(b, a \text{ live})$ (i.e. roles swapped), $U = P_1(\text{color 1 absent, live } a, b)$: total rate $a + b$;

$$
\begin{aligned}
(a+b)F &= a\left[\tfrac{1}{2}\cdot 0 + \tfrac{1}{2}U\right] + b\left[\tfrac{1}{2}\cdot 1 + \tfrac{1}{2}F\right],\\
(a+b)U &= a\left[\tfrac{1}{2}F + \tfrac{1}{2}\cdot 0\right] + b\left[\tfrac{1}{2}G + \tfrac{1}{2}\cdot 0\right],\\
(a+b)G &= b\left[\tfrac{1}{2}\cdot 0 + \tfrac{1}{2}U\right] + a\left[\tfrac{1}{2}\cdot 1 + \tfrac{1}{2}G\right].
\end{aligned}
$$

Solving the 3×3 system: from the first and third, $F = (aU + b)/(2a + b)$, $G = (bU + a)/(2b + a)$; substituting into the second gives $U\,[2(a+b) - a^2/(2a+b) - b^2/(2b+a)] = ab[1/(2a+b) + 1/(2b+a)]$. With $t = a+b$, $p = ab$: $(2a+b)(2b+a) = 2t^2 + p$; the bracketed left factor equals $3t(t^2+p)/(2t^2+p)$ and the right side $3pt/(2t^2+p)$, hence $U = p/(t^2+p) = ab/D$ since $t^2 + p = D$. Back-substitution and the factorization $2a^2 + 3ab + b^2 = (2a+b)(a+b)$ yield $F = b(a+b)/D$, $G = a(a+b)/D$. The sum check is $(a+b)^2 + ab = D$. □

Remark A.5.1 (A.5.1). Checks: $a = b \Rightarrow P_1 = P_2 = 2/5$, $P_3 = 1/5$; $a < b \Rightarrow P_1 > P_2$ (minority advantage persists); $b \to 0^+$ recovers Theorem A.4 limits.

Theorem A.6 (General Q: first-step linear system)**.** *Order states by number of live colors; consensus states are absorbing and every non-consensus state is transient (Lemma A.2 kills colors at positive rate until $L = 1$). Winning probabilities are the unique solution of the first-step system $(I - P_{TT})x = P_{TA}x_A$ over transient states T, where P_{TT} is substochastic with spectral radius < 1, hence $I - P_{TT}$ invertible; the recursion closes at Theorem A.5 for $Q = 3$ levels.*

Proof. Standard Markov-chain hitting-probability theory; invertibility follows from transience (Neumann series). □

Remark A.6.1 (A.6.1). This replaces any Schur-complement bookkeeping by an invertibility argument that cannot fail; explicit constants for $Q \geq 4$ are not claimed.

S2 (B). Coverage and ball geometry

Proposition B.1 (Ball counts, non-wrap)**.** *On the d-dimensional hypergrid with 3-consecutive-node hyperedges per axis, one hyperedge step moves ≤ 2 lattice sites per axis; the per-axis cost of offset o is $\lceil |o|/2 \rceil$; in the non-wrap regime $|B_r| = \sum_{c_1+\cdots+c_d \leq r} \prod_k w(c_k)$ with $w(0) = 1$, $w(c) = 4$. Hence $|B_r| = 4r + 1$ (1D) and $|B_r| = 8r^2 + 1$ (2D).*

Proof. Offsets of cost $c \geq 1$ are $\{\pm(2c-1), \pm 2c\}$ (4 of them); convolution over axes gives the count; the 2D evaluation $(1+4r)+\sum_{c=1}^{r} 4(1+4(r-c)) = 8r^2+1$ is direct. □

Proposition B.2 (1D exact: ball, R1, R2)**.** *On the 1D hypergrid: $d(i,j) = \lceil \delta(i,j)/2 \rceil$ with δ the circular lattice distance; $\beta = \min(n, 4r+1)$; and exactly (no asymptotic qualifier):* R1 $\iff$ $2\beta > n$; R2 $\iff$ $\beta = n$ *or* $3\beta \geq 2n - 1$ *($\iff$ $n \leq 6r + 2$ when $\beta = 4r + 1 < n$).*

Proof. A 3-node hyperedge step moves ± 1 or ± 2 sites, giving the distance formula and arc balls. R1 is $\beta > n/2$. For R2 with $\beta < n$: the residual is an arc of $n - \beta$ nodes, lattice diameter $n - \beta - 1$, hypergraph diameter $\lceil (n - \beta - 1)/2 \rceil$; R2 is $\lceil (n - \beta - 1)/2 \rceil \leq r$, i.e. $n - \beta - 1 \leq 2r$; with $r = (\beta - 1)/4$ this is $2n - 2\beta - 2 \leq \beta - 1 \iff 3\beta \geq 2n - 1$; substituting $\beta = 4r + 1$ gives $n \leq 6r + 2$. Since R1 gives $\beta > n/2$, the residual arc has length $n - \beta < n/2$, strictly less than the half-circumference; its hypergraph diameter is therefore computed by the circular lattice distance on the residual arc itself, with no wrap-around caveat. □

Remark B.2.1 (B.2.1). The finite-size equivalences above replace all "asymptotically $q > 2/3$" statements; asymptotics are corollaries.

Proposition B.3 (Coverage radius, non-wrap)**.** *For fixed $0 < q < 1$ and d, the minimal non-wrap radius achieving $|B_r| \geq qn$ satisfies $r_{\min} = \Theta(n^{1/d})$ with constants from Proposition B.1; counting-error shifts r by $O(1)$ since $|B_r|$ is strictly increasing in r.*

Proof. Volume monotonicity pins the radius between consecutive integers. □

Proposition B.4 (Wrapped regime; numerical evidence only)**.** *For fixed q on d-dimensional tori, $|B_r|/n \to$ rescaled diamond volume; lattice counts are Riemann sums with $O(n^{-1/d})$ boundary layers; monotonicity pins the radius.* Status: numerical evidence, not a complete proof. *The Riemann-sum and boundary-layer arguments are sketches; they are validated numerically but not promoted to a theorem. This statement is* ***not*** *used as a foundation for any containment or band result below; it appears only for completeness of the geometry section.*

Proposition B.5 (Configuration model, sparse; conditional theorem with sketch)**.** *If $(k-1)^{2r+1} = o(n)$ then, plain or simplicity-conditioned, $|B_r|/n \to$ the non-backtracking tree volume $/n$ with collision probability $o(1)$ (union bound over endpoint checks, each $O((k-1)^{2r}/n)$); hence Poisson-type corrections carry no extra content in the sparse regime; at the crossover the Poisson form is a* conjecture*, not a theorem.* Status: proof sketch. *The union-bound argument is not written out in full; the sparse-regime limit is used only as motivation for the truncated-ball protocol in Protocol F.4 and is not a hypothesis of any theorem below.*

S3 (C). Exact band at high coverage

Lemma C.0 (Entry and initial conditions)**.** *(i) From a monochrome state, the first fire at i creates minority R_i whenever $\beta_i > n/2$. (ii) From any strict-majority state, the next rotation epoch produces a minority $A \subseteq R_i$ for the firing node i. (iii) Tie states are left at the first event and never re-entered. (iv) For any initial configuration (including i.i.d. colorings), the embedded post-rotation chain enters a closed communicating class in finite time a.s. Different closed classes may support different stationary distributions; however, Lemma C.5 proves that the stationary mean minority fraction is the same on every closed class. Consequently, the stationary majority fraction in Theorem C.6 is independent of the initial law and of the choice of closed class.*

Proof. (i) The flipped class occupies B_i; since $\beta_i > n/2$ it is the new majority and the minority is R_i. (ii) is Lemma C.2. (iii) is Proposition C.1(tie). (iv) The epoch class is closed by Lemma C.2 and reached at the first rotation/birth epoch, which occurs a.s. in finite time on a finite state space. Stationary distributions of a finite Markov chain with several closed classes need not be unique. The class-independence used in Theorem C.6 is *not* a consequence of support on closed classes; it is proved separately in Lemma C.5. □

Remark C.0.1 (C.0.1). This lemma only justifies using stationary initial data for the observables in Theorem C.6. It does not assert uniqueness of the stationary distribution.

Proposition C.1 (Rotation identity, no equal-ball assumption)**.** *Assume $Q = 2$ and $\beta_i > n/2$ for every i. From a strict-majority state (M, A): a rotation occurs iff the firing node lies in M; the instantaneous rotation rate is $|M|$; at stationarity the rotation-event probability equals $E[M]$ (rates 1 per node, total rate n). Tie states carry zero stationary mass.*

Proof. Majority fire at i: new class sizes are $|A| + |M \cap B_i|$ vs $|M \setminus B_i|$; the margin is $|A| + |M \cap B_i| - |M \setminus B_i| = (2\beta_i - n) + 2|A \setminus B_i| > 0$ since $\beta_i > n/2$; hence the label swaps (rotation). Minority fire at i: margin of $M \cup (A \cap B_i)$ over $A \setminus B_i$ only increases; no rotation. Rate: each $i \in M$ fires at rate 1 and rotates. Stationary rotation probability $= E[|M|]/n = E[M]$. Tie: with $|M| = |A|$, the same margin formula (with $|M| - |A| = 0$) gives strict majority after any fire; strict-majority states map to strict-majority states, so ties are transient. □

Remark C.1.1 (C.1.1). In the majority-fire branch only nodes of the *majority* class inside B_i flip; minority nodes inside B_i already carry the target color and are unchanged. Any wording to the contrary is a labeling error and does not affect the margin algebra above.

Remark C.1.2 (C.1.2). Only $\beta_i > n/2$ per firing node is used; equal balls are **not** required for Proposition C.1. Equal balls enter first at Lemma C.3.

Lemma C.2 (Epoch invariant and self-containment)**.** *Under R1∧R2 with $Q = 2$, for every post-rotation minority A_m with firing node i_m: $A_m \subseteq R_{i_m}$; hence $A_m \subseteq B_j$ for all $j \in A_m$ ("self-containment"), and $|A_m| \leq n - \beta$.*

Proof. Base: Lemma C.0(i). Induction: given $A \subseteq R_i$, the next fire is uniform on V. Branch 1 ($j \in A$): self-containment (from R2 applied to

$R_i \ni j$: $R_i \subseteq B_j$, and $A \subseteq R_i$) gives $A \subseteq B_j$, so the minority is erased; the system is monochrome and the next fire creates $A' = R_{i'}$. Branch 2 ($i \in M$): by Proposition C.1 a rotation occurs and the new minority is $M \setminus B_i \subseteq R_i$. Self-containment of the new minority follows again from R2 on R_i. □

Lemma C.3 (Overlap identity; equal balls β)**.** *Under Lemma C.2, for i uniform in M: $E[|A \cap B_i| \mid a] = \frac{a(\beta-a)}{n-a}$.*

Proof. $\sum_{i\in M} |A \cap B_i| = \sum_{j\in A} |B_j \cap M|$. For $j \in A$, self-containment gives $B_j \cap A = A$, hence $|B_j \cap M| = \beta - a$. Sum $= a(\beta-a)$; divide by $|M| = n-a$. □

Lemma C.4 (Exact affine linearization)**.** $E[a_{m+1} \mid a_m = a] = (1-q)(n-a)$.

Proof. Branch 1 (prob. a/n): $a' = n - \beta$ (Lemma C.2). Branch 2 (prob. $1 - a/n$): $a' = n - \beta - a + |A \cap B_i|$, mean $n - \beta - a + \frac{a(\beta-a)}{n-a} = \frac{(n-\beta)(n-2a)}{n-a}$ after combining over $n-a$. Weighting: $\frac{a}{n}(n-\beta) + \frac{n-a}{n} \cdot \frac{(n-\beta)(n-2a)}{n-a} = (n-\beta)\frac{a+n-2a}{n} = (1-q)(n-a)$. Affine in $a \Rightarrow$ no Jensen gap under any law. □

Lemma C.5 (Stationary epoch mean and cycle structure)**.** *Let π be any invariant distribution of the finite epoch chain; $\bar{x} = E_\pi[a]/n$. Then $\bar{x} = \frac{1-q}{2-q}$, independent of π; and $P(K = 2 \mid a) = a/n$, so $E[K] = 1 + \bar{x}$.*

Proof. Take π-expectations in Lemma C.4: $\bar{x}n = (1-q)(n - \bar{x}n) \Rightarrow \bar{x} = (1-q)/(2-q)$; the linear equation has a unique solution, hence independence of π (and of the closed class supporting π). A cycle has a second fire iff the first fire is a minority fire, probability a/n given the epoch state. □

Theorem C.6 (Exact band curve)**.** *Assume $Q = 2$, R1, R2, equal balls β. Then $E[M] = \frac{2-q}{3-2q}$.*

Proof. Start the epoch chain in an invariant π (exists: finite state space). The marked cycle process (Y_m, τ_m) (reward $Y = \int M\, dt$ over cycle m, length τ) is then *stationary* (cycles are functionals of a stationary chain); they are **not** independent, and no independence is used. By the Birkhoff ergodic theorem for stationary sequences, a.s. $\frac{1}{M}\sum_{m\le M} Y_m \to E[Y \mid \mathcal{I}]$ and $\frac{1}{M}\sum \tau_m \to E[\tau \mid \mathcal{I}]$, $\mathcal{I}$ the invariant σ-field; since $E[\tau \mid \mathcal{I}] \ge 1/n > 0$, the ratio converges a.s. to $E[Y \mid \mathcal{I}]/E[\tau \mid \mathcal{I}]$. Conditioning on the closed class entered, these conditional expectations equal the class-wise means $E_c[Y], E_c[\tau]$. Compute them: A cycle starts immediately after a rotation and ends at the next rotation. Within the first gap (mean $1/n$), the state is non-monochrome;

if the first fire is a minority fire (probability a_m/n given the epoch state), the minority is erased and the subsequent fire (mean gap $1/n$) recreates a monochrome-then-newborn state, giving $K = 2$; otherwise the first fire is a majority fire that rotates the label, giving $K = 1$. Thus $K \in \{1, 2\}$ with $P(K = 2 \mid a_m) = a_m/n$. Gaps are Exp (mean $1/n$), independent of colors; during the first gap $M = (n-a)/n$, during the optional second (monochrome) gap $M = 1$; hence $E_c[Y] = \frac{1}{n}E_c[1-a/n] + \frac{1}{n}P_c(K = 2) = \frac{1}{n}(1-\bar{x}_c) + \frac{1}{n}\bar{x}_c = \frac{1}{n}$ and $E_c[\tau] = \frac{1}{n}(1 + \bar{x}_c)$, where $\bar{x}_c$ is the class-wise mean; Lemma C.5 gives $\bar{x}_c = \bar{x}$ for every class. Indeed, since π is supported on a closed class, the restriction of the epoch chain to that class is itself a finite Markov chain on a finite state space; applying Lemma C.5 to the restricted chain gives the class-wise mean $\bar{x}_c = (1-q)/(2-q)$, independent of the choice of class. Therefore the ratio equals $\frac{1}{1+\bar{x}}$ on every class, hence a.s. overall. Substituting $\bar{x} = \frac{1-q}{2-q}$: $E[M] = \frac{1}{1+\bar{x}} = \frac{2-q}{3-2q}$. □

Remark C.6.1 (C.6.1). The renewal–reward step uses stationarity + Birkhoff + class-wise constancy; the phrase "i.i.d. cycles" never appears and is not needed. Reducibility of the epoch chain is explicitly handled.

Corollary C.7 (Threshold location; 2D conditional)**.** *$f(q) = (2-q)/(3-2q)$ is strictly increasing ($f' = \frac{1}{(3-2q)^2} > 0$), $f(1/2) = 3/4$, $f(2/3) = 4/5$; hence any band-predicted crossing of $M = 3/4$ lies at $q^*_{\text{band}} = 1/2$, below the exact-regime domain edges. For 2D hypergrids the corollary applies **only where R1∧R2 holds**; at the finite sizes used, R1∧R2 was verified by exhaustive BFS (e.g. linear size $L = 12$, $n = L^2 = 144$, at $\beta/n \approx 0.771$; $L = 30$, $n = 900$, at $\beta/n \approx 0.777$); the value $7/9$ is a numerical observation, **not** a theorem, and no 2D threshold theorem is claimed.*

Remark C.7.1 (C.7.1). Any operational threshold q_{dyn} defined by interpolating measured $M(q)$ at a level (e.g. 3/4) is a *protocol observable*; statements about it require the measurement protocol and empirical monotonicity, and are labeled Observation, never Theorem.

Theorem C.8 (Dwell two-point law)**.** *Under Theorem C.6: $E[K] = 1 + \bar{x} = \frac{1}{E[M]}$, $E[T] = \frac{1+\bar{x}}{n}$, and $\text{CV}^2_T = \frac{7-8q+2q^2}{(3-2q)^2}$.*

Proof. $K \in \{1, 2\}$ with $P(K = 2) = \bar{x}$ (Lemma C.5); $\tau = G_1 + BG_2$ with $G_1, G_2 \sim \text{Exp}(n)$ (mean $1/n$), $B \sim$Bernoulli($\bar{x}$). The G's are independent of the color process (exponential clocks are memoryless and independent of the

discrete event sequence), and B is a functional of the color state at the first gap, hence independent of G_2; therefore $E[\tau] = (1+\bar{x})/n$; $\mathrm{Var}(\tau) = \frac{1}{n^2} + \frac{\bar{x}}{n^2} + \frac{\bar{x}(1-\bar{x})}{n^2} = \frac{1+2\bar{x}-\bar{x}^2}{n^2}$. With $\bar{x} = \frac{1-q}{2-q}$: $1+\bar{x} = \frac{3-2q}{2-q}$ and $1+2\bar{x}-\bar{x}^2 = \frac{7-8q+2q^2}{(2-q)^2}$, giving $\mathrm{CV}^2 = \frac{7-8q+2q^2}{(3-2q)^2}$. $E[K] = 1+\bar{x} = 1/E[M]$ follows from $E[M] = 1/(1+\bar{x})$. □

S4 (D). Containment theory

Proposition D.1 (c_G on vertex-transitive and configuration geometries)**.** *With i, j independent uniform: $c_G = E_{i,j}\mathbf{1}\{R_i \subseteq B_j\}$. On vertex-transitive H with uniform β: $c_G = \frac{1}{n} \cdot |\{j : R_i \subseteq B_j\}|$ independent of i, i.e. c_G equals the per-residual containment count over n. On configuration models, $c_G = E_i[\,|\{j : R_i \subseteq B_j\}|/n\,]$. This is an ensemble-averaged overlap count. It is* not *determined by ball-size dispersion alone: two graphs with the same degree sequence, or even the same ball-size distribution, can have different c_G because the overlap pattern of balls and residuals differs. All values must be computed by BFS/subset enumeration; no inference from β alone is valid.*

Proof. Exchange of expectations and symmetry of d give the stated forms; vertex-transitivity makes the inner count i-independent. □

Proposition D.2 (Necessary condition for containment)**.** *If $R_i \subseteq B_j$ then $\mathrm{diam}_H(R_i) \leq 2r$; hence on families with radius r, $\tilde{c} > 0$ requires some residual of diameter $\leq 2r$. All values must be computed by BFS; no inference from β alone is valid.*

Proof. Triangle inequality + supremum over R_i. □

Proposition D.3 (R2 ⇒ certain extinction; non-uniform version)**.** *Under R2: for every i, $\{j \in R_i : R_i \subseteq B_j\} = R_i$, hence the per-residual conditional probability is 1 and $\tilde{c} = 1$; moreover $c_G = \frac{1}{n^2}\sum_i |\{j : R_i \subseteq B_j\}| \geq \frac{1}{n^2}\sum_i |R_i|$. If balls are uniform, $c_G = (2\beta - n + 1)/n$ in 1D (Proposition E.1); if not, the per-ball formula uses β_i.*

Proof. R2 gives $R_i \subseteq B_j$ for all $j \in R_i$; the rest is definition. □

Remark D.3.1 (D.3.1). $\tilde{c} = 1$ throughout the exact regime is a *consistency requirement* on any extinction theory; it is not evidence that $\tilde{c}$ mediates intermediate-regime effects (main text Sec. VI).

Proposition D.4 (Intersection reformulation, equal balls)**.** $R_i \subseteq B_j \iff B_i \cup B_j = V \iff |B_i \cap B_j| = 2\beta - n$*; necessary condition* $2\beta \geq n$*; if* $2\beta < n$ *then* $c_G = 0$*.*

Proof. Inclusion–exclusion; union $\leq n$. □

S5 (E). Exact 1D containment and regions

Proposition E.1 (c_G, exact)**.** *On 1D with* $\beta = \min(n, 4r+1)$*:* $c_G = \min\left(1, \frac{\max(0, 2\beta - n + 1)}{n}\right)$*. In particular at* $2\beta = n$ *the residual is itself a ball and* $c_G = 1/n \neq 0$*.*

Proof. Residual arc length $n - \beta$; a ball arc of length β contains it iff its (unique, β odd) center lies in an interval of $2\beta - n + 1$ positions when positive; divide by n centers; cap at 1 when $\beta = n$. At $2\beta = n$ the count is 1, giving $1/n$. □

Proposition E.2 ($\tilde{c}$, exact, three regimes)**.** $\tilde{c} = 0$ *if* $2\beta < n$*;* $\tilde{c} = \frac{2\beta - n + 1}{n - \beta}$ *if* $2\beta \geq n$ *and* $3\beta < 2n - 1$ *(value* $1/\beta$ *at* $2\beta = n$*);* $\tilde{c} = 1$ *if* $\beta \geq \frac{2n-1}{3}$ *(each case with the* min / max *convention at the endpoints). The function has a jump* $0 \to 1/\beta$ *at* $2\beta = n$ *(a finite-size effect) and is continuous at the upper boundary* $3\beta = 2n - 1$*.*

Proof. Admissible firing centers $j \in R$ with $R \subseteq B_j$ form an interval of length $2\beta - n + 1$; intersect with R (length $n - \beta$). If $2\beta < n$, no ball contains R. If $2\beta = n$, then $|B_j| = |R| = \beta$ and containment forces $B_j = R$; the unique center of R (odd β) lies in R, giving exactly one admissible firing node. If $2\beta > n$ and $3\beta < 2n - 1$, the admissible interval has length $2\beta - n + 1$ and lies strictly inside R. If $3\beta \geq 2n - 1$ the interval covers R (ratio 1). Continuity holds at $3\beta = 2n - 1$. □

Remark E.2.1 (E.2.1). The denominator is $n - \beta$ (residual size), not β and not an asymptotic $(1 - q)$ form; the $+1$ is retained (finite-size exact).

Proposition E.3 (Three dynamical regions)**.** *Region I:* $3\beta \geq 2n - 1$ *(R1∧R2, band applies,* $\tilde{c} = 1$*); Region II:* $n/2 < \beta < (2n-1)/3$ *(R1 strict for* $\beta > (n+1)/2$*, R2 false,* $0 < \tilde{c} < 1$*); Boundary case:* $2\beta = n$ *(*$c_G = 1/n$*,* $\tilde{c} = 1/\beta$*); Region III:* $2\beta < n$ *(R1 false,* $\tilde{c} = 0$*). Asymptotics* $q \geq 2/3$*,* $1/2 < q < 2/3$*,* $q \leq 1/2$ *are corollaries.*

Proof. Combine Proposition B.2 with Propositions E.1–E.2. □

Validation E.4.. Brute-force bitmask enumeration of all ball pairs for every $n \in [12, 40]$ and all r, *including the boundary case* $2\beta = n$ *at* $(n, r) = (18, 2)$ *where* $c_G = 1/18$ *and* $\tilde{c} = 1/9$: max $|\text{enum} - \text{formula}| < 10^{-12}$ for both c_G and $\tilde{c}$. Suite ships with the code release.

S6 (F). Protocols, validation ledger, and disclosures

Proposition F.1 (Newborn identity; protocol theorem)**.** *From monochrome, the first fire at i creates minority* R_i*; conditioning the next event on a minority fire, the firing node is uniform in* R_i *and (with* $Q = 2$*) the target is the majority color; extinction occurs iff* $R_i \subseteq B_j$*; hence the conditional extinction probability is* $|\{j \in R_i : R_i \subseteq B_j\}|/|R_i|$*, and averaging over i (weight* $|R_i|$*) gives* $\tilde{c}$*.*

Proof. Direct from the flip rule with $Q = 2$. □

Remark F.1.1 (F.1.1 Monte Carlo validation). An initial implementation of the newborn-identity Monte-Carlo test contained a bookkeeping defect: the newborn minority tracker was not reset after majority-fire rotations, causing the chain to stall in the post-rotation regime and returning zero successes across all protocol cells. The defect was identified and fixed by explicitly tracking monochrome episodes and clearing the tracker on any subsequent event (minority fire records a trial; majority fire records nothing). The corrected simulator yields conditional extinction probabilities matching the geometric $\tilde{c}$ within binomial error (e.g. $n = 15$, $r = 2$: 29,489 trials, MC 0.6664 ± 0.0027 vs. $\tilde{c} = 2/3$; $n = 25$, $r = 3$: 19,535 trials, MC 0.1666 ± 0.0027 vs. $\tilde{c} = 1/6$). Proposition F.1 is thereby empirically validated.

Proposition F.2 (BFS containment by popcount)**.** *For equal balls,* $R_i \subseteq B_j \iff \text{popcount}(B_i \,\&\, B_j) = 2\beta - n$*; the* $O(n^2)$ *algorithm (target* $= 2\beta - n$*; if negative return 0) is exact.*

Proof. Proposition D.4. □

Protocol F.3 (Unit tests and solver specification)**.** (i) *Static:* Propositions E.1–E.2 vs brute force, $n \in [12, 40]$: **PASSED** ($< 10^{-12}$).

(ii) *Exact chain solver:* two implementations are shipped, and the manuscript's Table S2 uses the first:

(a) *Reachable-orbit solver with dihedral quotient* (`gvm_validation_suite.py`, tasks T1 and T5). States are orbits of reachable minority sets under the

dihedral group D_n; the orbit BFS is built by applying the fire rule to the orbit representative and quotienting the result. The state space is therefore much smaller than 2^{n-1}; reported orbit counts are $8, 24, 80, 80, 288, 1088$ for $n = 9, 15, 17, 21, 23, 25$ (Table S2). Transition matrix: row-stochastic, entries $1/n$ per firing node; stationary distribution by lazy power iteration $(\frac{1}{2}I + \frac{1}{2}P)$ from the monochrome orbit; convergence monitored by the L^1 residual $\|\pi P - \pi\|_1$, reported as $\leq 1.4 \times 10^{-14}$.

(b) *Full-enumeration solver with color symmetry* (`src/exact/exact_chain_1d.py`). Enumerates all states with minority size $\leq (n-1)/2$ (color-complement canonicalization only, no dihedral quotient); used as an independent cross-check for $n \leq 21$ and not used to generate any published table. Its state counts appear in Table S3.

The description in the main text ("reachable subspace and dihedral quotient") refers to (a). Implementation (b) is documented here only to prevent confusion if the repository is inspected directly.

(iii) *Simulator cross-check targets (specified; execution status in ledger):* native 1D balls at $(n, r) = (15, 2), (17, 2), (21, 3)$ must reproduce $M = 0.773810, 0.741294, 0.781711$ within ± 0.002; Region I: 1D $n = 400$, $q \approx 0.75$ vs band 0.83333 ± 0.003; 2D $L = 30$ at $\beta/n \approx 0.777$ (BFS-verified R1$\wedge$R2) vs band $\approx 0.845 \pm 0.005$.

(iv) *Newborn MC:* validated per Remark F.1.1.

Protocol F.4 (Controlled regression test of the containment hypothesis (v1 pre-registered; v2 post-hoc re-evaluation))**.** Design as main-text Protocol 1: families {1D cycle, 2D torus king-adjacency, configuration $k = 6$, Watts–Strogatz $k = 6, p = 0.1$}; $n \in \{100, 144, 400\}$; $q \in \{0.55, \ldots, 0.80\}$; truncated balls $\beta = \text{round}(qn)$ by BFS layering (this matches the configuration `configs/v2.yaml` and `measure_pipeline.py`); $\tilde{c}$ by subset enumeration; $Q = 2$ simulation 2×10^5 events, 5×10^4 burn-in, batch means 2000 $\Rightarrow$ $M \pm$SE; weighted fits $M \sim q + C(\text{family}) + C(n)$ (reduced) vs $M \sim q + \tilde{c} + C(\text{family}) + C(n)$ (full), weights $1/\text{SE}^2$.

Data structure (normative). The file `measurements_v2.csv` contains 216 seed-level rows (4 families $\times$ 3 sizes $\times$ 6 coverages $\times$ 3 seeds). Rows sharing a (family, n) pair share a single quenched graph realization; the three seeds vary only the dynamics randomness. The regression is fit to these 216 rows.

Graph realizations (normative). Each family is drawn once per (family, n) pair with a fixed seed. The configuration-model family is drawn as a random multigraph and projected to a simple graph by removing self-loops and

merging multi-edges; the resulting graph is not degree-regular, and the family offsets in Table S4 are properties of this specific realization, not of the full configuration-model ensemble.

Bootstrap (normative). The primary inference uses a two-level cluster bootstrap: level 1 resamples the 12 (family, n) clusters with replacement; level 2 resamples the 18 seed-level rows within each selected cluster with replacement. $B = 500$, seed 9999, percentile CIs. The iid bootstrap of the initial v1 analysis treats the 216 rows as independent and is retained as a sensitivity check only; it underestimates variance because it ignores the shared graph realizations (see S6.5).

Family-effect norm (normative). The reference-invariant norm is the L^2-norm of the three pairwise family contrasts (2D − WS, 2D − config, WS − config). The reduction statistic is $(\|\text{contrasts}_{\text{reduced}}\| - \|\text{contrasts}_{\text{full}}\|)/\|\text{contrasts}_{\text{reduced}}\|$. The initial v1 analysis used the non-invariant norm of family coefficients relative to the 1D reference; that statistic is not used in the v2 re-evaluation.

Acceptance (fixed in advance): (a) γ significant and positive; (b) reference-invariant family-effect norm reduced $\geq 30\%$.

Result (cluster bootstrap, primary). $\hat{\gamma} = -0.0165$; bootstrap 95% CI $[-0.0230, +0.0045]$; family-norm reduction median -0.031 (95% CI $[-0.189, +0.075]$); fraction of samples with reduction $\geq 30\%$: 0.002.

Result (iid bootstrap, sensitivity). $\hat{\gamma} = -0.0179$; bootstrap 95% CI $[-0.0238, -0.0125]$; family-norm reduction median -0.865 (95% CI $[-1.352, -0.530]$). The last number uses the non-invariant reference-relative norm of v1 and is not directly comparable to the invariant reduction above.

Decision. Under the primary cluster bootstrap, both acceptance criteria fail but in the direction of the null: γ is negative yet indistinguishable from zero, and the family-effect norm is unchanged. Under iid bootstrap, γ is nominally significant and the norm appears inflated, but both signals are artifacts of ignoring the cluster dependence and of using a reference-dependent norm. We report the cluster result as primary.

Interpretation. Because v2 changed both the bootstrap scheme and the family-effect norm definition after inspecting the data, the result is exploratory evidence against the one-step mediation hypothesis, not a confirmatory falsification of the hypothesis in general. We do not claim that one-step containment plays no role in the intermediate regime.

Limitations (normative): (1) $\tilde{c}$ describes only the first fire of a newborn from monochrome; (2) it does not describe shrunken or fragmented minorities;

(3) with four families, $\tilde{c}$ variation concentrates in 1D, limiting power; a continuous geometric family would strengthen future tests; (4) criteria failure $\Rightarrow$ no mechanism claim; (5) only 12 clusters are available for the primary bootstrap, which limits the precision of the cluster-level CI.

S6.5 (F.5). Cluster dependence in quenched-hypergraph simulations

A common practice in simulation-based studies of quenched hypergraph dynamics is to resample seed-level runs as if they were independent. This is incorrect whenever multiple seeds share the same graph realization, as they do in any quenched-geometry protocol: the seeds vary only the dynamics randomness, not the geometry. Ignoring this dependence inflates the effective sample size and produces anti-conservative confidence intervals.

We encountered exactly this artifact in the v1 analysis. The file `measurements_v2.csv` contains 216 seed-level rows; the three seeds sharing a (family, n) pair share the same quenched graph. Under iid resampling of all 216 rows, the $\tilde{c}$ coefficient was estimated at $\hat{\gamma} = -0.0179$ with 95% CI $[-0.0238, -0.0125]$. Under the appropriate two-level cluster bootstrap (resampling the 12 (family, n) clusters first, then the 18 rows within each), the same point estimate yields 95% CI $[-0.0230, +0.0045]$ and is not distinguishable from zero at the cluster level.

A second, independent artifact appeared in the family-effect norm. The v1 analysis measured the norm of family coefficients relative to the 1D reference family, giving a median reduction of -0.865 (apparent inflation) under $\tilde{c}$ control. This norm is not invariant under reparametrization: changing which family is the reference changes the norm and hence the reduction. The reference-invariant L^2-norm of pairwise family contrasts is the appropriate statistic; under the same $\tilde{c}$ control it gives a median reduction of -0.031 (95% CI $[-0.189, +0.075]$), indistinguishable from zero.

We flag both artifacts explicitly because they are likely present in other simulation studies of quenched hypergraph models. Two remedies are simple: use a two-level cluster bootstrap whenever multiple seeds share a graph realization, and use pairwise-contrast norms whenever the reference family is arbitrary.

Validation ledger (status at this release).. Normative note: the main-text sentence asserting simulator cross-checks must, at submission, match this ledger (either report executed values or state "targets specified in the code release").

Table S1: Validation ledger.

Item	Status	Evidence
Static 1D containment vs brute force	PASSED	E.4
Exact-chain stationary residuals	PASSED ($\leq 1.4 \times 10^{-14}$)	Table S2
H3 exhaustive small-n	PASSED (0 violations, 6 cases up to $n = 25$)	Table S2
Simulator vs exact chain (1D Region II)	PASSED (± 0.002)	code release
Simulator vs band (Region I, 1D & 2D)	PASSED ($\lvert\Delta\rvert < 0.003$)	code release
Newborn MC (F.1)	PASSED (defect fixed, matches $\tilde{c}$)	Remark F.1.1
Regression/bootstrap (F.4)	EXECUTED — both acceptance criteria fail; cluster-primary result is a null result	Table S4; S6.5

S7 (G). Conjectures and open problems

Conjecture G.1 (H3)**.** *In Region II, $M(q) \leq f(q)$ for all n, r.*

Evidence. Exact solutions at odd $n \leq 25$ (six cases; deviations from -0.0021 to -0.0184; Table S2). Reconstructed rationals $43/57$, $65/84$, $149/201$, $265/339$, $72/95$, $319/433$ (denominators $3 \cdot 19$, $2^2 \cdot 3 \cdot 7$, $3 \cdot 67$, $3 \cdot 113$, $5 \cdot 19$, 433) show no simple universal form; no coupling/monotonicity proof known; **not** promoted to a theorem.

Conjecture G.2 (Scaling near threshold)**.** *Geometric scaling form near q^* on d-dimensional hypergrids; untested; no coefficients quoted.*

Conjecture G.3 (Collapse by $\tilde{c}$: not supported within the tested protocol)**.** *The collapse hypothesis was accepted only if Protocol F.4 criteria held; the protocol does not support it within its tested range.*

Open G.4 (Config–2D gap)**.** Candidates: residual ball-size heterogeneity (partially controlled by truncation), expander structure, fractional-radius interpolation; required test: size-truncated balls separating heterogeneity from geometry (executed in Protocol F.4; gap persists as family intercepts).

Heuristic G.5 (Fragmentation)**.** Below R2 minorities may fragment; no evidence for a low-dimensional closed Markov description on shape variables; a heuristic, not a no-go theorem.

S7.1 Limitations and scope of the results

We state the boundaries of what the present paper establishes, so that the claims are read at the correct strength.

1. *Regimes without exact analytical characterization.* The band (Theorem C.6) is proven only under R1∧R2 with equal ball sizes. Region II ($n/2 < \beta < (2n-1)/3$) and Region III ($2\beta < n$) have no closed-form stationary characterization here; Conjecture G.1 is numerically supported, not proven.
2. *Results with numerical evidence only.* Propositions B.4 (wrapped torus) and B.5 (sparse configuration model) are stated with sketches and validated numerically; they are not used as hypotheses of any theorem. Table S2 contains numerical stationary solutions; the "rational" column is a numerical reconstruction, not an algebraic certificate.
3. *Single graph realization per* $(family, n)$. The regression in Protocol F.4 uses one fixed graph per (family, n) pair; the 3 seeds vary only dynamics randomness. The family offsets in Table S4 are therefore properties of this specific realization set, not of the full graph ensemble.
4. *Limited family coverage.* Four families (1D cycle, 2D torus, configuration $k = 6$, Watts–Strogatz $k = 6, p = 0.1$) and three sizes; $\tilde{c}$ variation concentrates in the 1D family, limiting statistical power for the mediation test (Protocol F.4, limitation 3).
5. *Interpretation of fragmentation.* The Region-II observation $M_{\text{exact}} < M_{\text{band}}$ is consistent with multi-step minority survival, but no theorem identifies fragmentation as the mechanism; the "fragmented survivor" language is interpretive, not a mathematical statement (Heuristic G.5).
6. *Equal-ball assumption.* The band result requires $\beta_i \equiv \beta$. Extension to unequal ball sizes is not covered here; Propositions C.1 and D.3 are the only statements that do not require equal balls.
7. *2-section reduction.* As stated in the graph-reduction remark of the main text (Remark 1), the dynamics depends on the hypergraph only through its 2-section graph. The paper characterizes quenched geometric group-flip dynamics on hypergraph-induced structures; it does not claim

Table S2: Exact chain, Region II (exhaustive odd $n \leq 25$). Solver residuals $\leq 1.4 \times 10^{-14}$; "orbits" = equivalence classes of minority sets under the dihedral group D_n. The "rational" column is a *numerical reconstruction* from the floating-point stationary solution; it is not an algebraic certificate. Exact rational verification, where reported, used a rational Gauss–Jordan solver; for the larger cases only the numerical stationary solution is available.

n	r	β	q	c_G	$\tilde{c}$	M_{exact}	M_{band}	diff	rational	orbits
9	1	5	0.5556	0.2222222	0.5000000	0.7543860	0.7647059	−0.0103199	43/57	8
15	2	9	0.6000	0.2666667	0.6666667	0.7738095	0.7777778	−0.0039683	65/84	24
17	2	9	0.5294	0.1176471	0.2500000	0.7412935	0.7575758	−0.0162822	149/201	80
21	3	13	0.6190	0.2857143	0.7500000	0.7817109	0.7837838	−0.0020729	265/339	80
23	3	13	0.5652	0.1739130	0.4000000	0.7578947	0.7674419	−0.0095471	72/95	288
25	3	13	0.5200	0.0800000	0.1666667	0.7367206	0.7551020	−0.0183815	319/433	1088

a higher-order interaction mechanism independent of the 2-section topology.

8. *Cluster-level inference.* The primary bootstrap resamples the 12 (family, n) clusters; this is the correct scheme given the shared graph realizations but yields wide cluster-level confidence intervals (e.g. γ 95% CI $[-0.0230, +0.0045]$). The iid bootstrap reported in the v1 analysis is retained only as a sensitivity check; see S6.5 for the methodological note.
9. *Cell-level standard errors (V2 release).* The aggregation stored in `measurements_v2.csv` contains only the arithmetic mean of seed-level standard errors, not the between-seed variance. The proper cell-level SE $\widehat{\mathrm{SE}}^2_{\text{cell}} = \frac{1}{R^2}\sum_r s_r^2 + \frac{1}{R}\widehat{\mathrm{Var}}_{\text{between}}(M_r)$ requires the second term, which the released CSV does not preserve. Because every cell contains exactly $R = 3$ seeds, the fallback weights differ from the proper weights by a single global factor; WLS point estimates and bootstrap percentile CIs are therefore unaffected, while analytic standard errors are conservative by approximately $\sqrt{3}$. All reported inference uses bootstrap CIs, not analytic SEs. Regenerating the aggregation with `M_std` and `n_runs` is the recommended fix for any expansion.

S8 (H). Numerical validation summary

S9 (I). Numerical appendix

Reproducibility.. Code (Numba BFS/containment, exact-chain solver, simulator, regression/bootstrap) and raw data (`measurements_v2.csv`,

Table S3: Exact chain, Region II, full-enumeration cross-check solver (no dihedral quotient). Only $n \in \{15, 17, 21\}$ are shown because the full state space 2^{n-1} exceeds the memory budget for $n = 23, 25$; the six-case exhaustive table is Table S2, which uses the reachable-orbit solver. Solver residuals $\leq 9.9 \times 10^{-13}$; "rational" column as in Table S2.

n	r	β	q	c_G	$\bar{c}$	M_{exact}	M_{band}	diff	rational	states	iters	L^1 resid
15	2	9	0.6000	0.2666667	0.6666667	0.7738095238	0.7777777778	−0.0039683	65/84	16 384	65	8.3×10^{-13}
17	2	9	0.5294	0.1176471	0.2500000	0.7412935323	0.7575757576	−0.0162822	149/201	65 536	75	9.9×10^{-13}
21	3	13	0.6190	0.2857143	0.7500000	0.7817109145	0.7837837838	−0.0020729	265/339	1 048 576	65	8.7×10^{-13}

Table S4: Regression, Protocol F.4. Point estimates and bootstrap inference under the two schemes. Primary = two-level cluster bootstrap ($B = 500$, 12 clusters of 18 rows). Sensitivity = iid bootstrap ($B = 500$, 216 rows). The family-norm reduction in the iid column uses the non-invariant reference-relative norm of v1; the primary reduction uses the invariant pairwise-contrast norm of v2. Negative values = inflation.

Term	intercept	q	$\tilde{c}$ (γ)	family
Point estimate	0.4804028	0.4999197	−0.0178670	
γ, 95% CI				
iid (sensitivity)			[−0.02380, −0.01255]	
cluster (primary)			[−0.02302, +0.00451]	
Family-norm reduction, median [95% CI]				
iid + non-invariant				−0.865 [−1.352, −0.530]
cluster + invariant				−0.031 [−0.189, +0.075]
Fraction of samples $\geq 30\%$ reduction: iid 0.000, cluster 0.002.				

`bootstrap.csv`, `sensitivity.csv`, `exact_1d_v2.json`, `newborn_test_v2.csv`) at https://github.com/phmldtngyn/Group-flip-consensus-on-hypergraphs. The primary cluster-bootstrap results are in `sensitivity.csv`; the iid sensitivity results are in `bootstrap.csv`. An earlier version of measurement with mislabeled coverage of native balls (ball size vs. degree) was discarded; no reported quantity uses it.